

BIANCA AZARTASH-NAMIN

AI Literacy Project Librarian, M.S. Physics

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Portfolio

Research & Professional Interests

AI Literacy Project Librarian at the University of Oklahoma Libraries, translating data services and AI into practical guidance for faculty, researchers, and students through instruction, consultations, and cross-unit collaboration. My work sits at the intersection of library science and emerging technology, spanning the research lifecycle, AI literacy and ethics, and academic computational infrastructure. An astrophysics background grounds my analytical approach to technical problem-solving and drives ongoing curiosity about emerging technology.

Education

University of Oklahoma

Master of Science in Physics

Norman, OK

Aug. 2022 - Dec. 2024

University of Oklahoma

Bachelor of Science in Astrophysics

Norman, OK

Jan. 2019 - Dec. 2021

- **Awards:** Homer L. Dodge Department of Physics and Astronomy Award for Meritorious Scholarship in Astrophysics

University of Oklahoma Libraries

AI Literacy Project Librarian

Norman, OK

July 2025 - Present

- **Consult one-on-one** with faculty, staff, researchers, and students on AI tools, research workflows, and digital scholarship resources across Norman, Health, and Tulsa campuses; listen to researcher needs; identify the right tool or resource; and refer to specialized services such as Digital Scholarship and Data Services (DSDS), OU Supercomputing Center for Education & Research (OSCER), or Data Institute for Societal Challenges (DISC) when applicable.
- **Lead coordination efforts** and formalize a working partnership between University Libraries, the Center for Faculty Excellence (CFE), and the Data Institute for Societal Challenges (DISC), building a cross-unit relationship that connects researchers with the right campus partners.
- **Design and deliver** technical and ethics-focused AI workshops to faculty, staff, researchers, and students in adherence to the ACRL CHOICE AI Literacy Framework and the library's AI Action Team Objectives, Key Results, and Actions (OKRAs) for research support, student success, and the broader UL Strategic Plan and One OU Libraries Initiative.
- **Develop digital learning materials** for University Libraries and Generative AI output verification tutorials that are in use by faculty for individual courses and required for first-year students at the University of Oklahoma.
- **Completed Carpentries Instructor training** and certification to teach software and data Carpentry workshops that make programming and reproducible workflows approachable for learners with no prior technical background.
- **Consult with faculty** on using generative AI for digital accessibility and develop a tool-kit grounded in WCAG 2.1 requirements to audit and remediate math equations in LaTeX, PDF, and other course content in Learning Management Systems (LMS).
- **Write instruction guides** and facilitate training for instruction and liaison librarians to integrate AI literacy into their own instruction and outreach, improving AI instruction statistics and supporting future assessment initiatives.
- **Represent University Libraries** across multiple campus working groups and committees, including the AI Action Team for Research Support and Student Success, the AI Education Working Group, and the UL Research Workshops Committee, building relationships and shared practice across all OU campuses.
- **Support instruction and liaison librarians** with AI features in library databases and workflows; meet with product executives and faculty to discuss AI features and tools; have delivered classroom instruction on critical AI concepts to over 20 classes, student organizations, and departments across campus to date.

Metropolitan Library System

Library Aide

Edmond, OK

April 2025 - July 2025

- **Provided reference and technology support** to a diverse patron base, connecting guests with library programs, digital services, and community resources.

Outlier

AI Training

Remote

Dec. 2024 - Jan. 2025

- **Developed use cases and prompts** tailored to academic and professional contexts, evaluating AI tool performance and providing feedback to improve factual accuracy and inclusive language.

Research Experience

University of Oklahoma

Graduate Research Assistant

Norman, OK

Aug. 2021 - Dec. 2024

- **Trained and mentored peers** in programming, technical/software systems, and research workflows, building shared understanding of research goals across the group.
- **Wrote data management plans** for NSF Graduate Research Fellowship Program (GRFP) and Future Investigators in NASA Earth and Space Science and Technology (FINESST) fellowship proposals, translating research plans into clear, funder-facing documentation and building familiarity with the research lifecycle.
- **Conducted computational data analysis** using OSCER's high-performance research computing infrastructure, producing reproducible programming scripts and data visualizations for peer-reviewed publications.

N-Body Shop

Jan. 2022 - Dec. 2024

- Analyzed bursty star formation and dark matter density profiles of dwarf galaxies using high-resolution cosmological simulations.

JWST (James Webb Space Telescope)

May 2024 - Dec. 2024

- Performed data reduction and visualization of white dwarf spectroscopy using JWST survey targets, with extensive handling of NIRSpec and MIRI data via the Mikulski Archive for Space Telescopes (MAST).
- Optimized the JWST data reduction pipeline for spectroscopic analysis workflows.

CERN (European Council for Nuclear Research) & ATLAS (A Toroidal LHC ApparatuS)

2019 - 2022

- Developed Python-based data analysis and visualization pipelines to process experimental sensor data from ATLAS pixel detector thermal shock testing.
- Designed and deployed the underlying data collection system: Arduino sensors for temperature/humidity, custom dew point calculations, and 3D-printed hardware for structured, repeatable sensor deployment.
- Investigated sensitivity of a search for exotic Higgs boson decays to long-lived particles (LLPs) decaying to tau lepton pairs, simulating signal events and processing them through the ATLAS Athena reconstruction framework and ROOT-based selection analysis.

Snowmass Early Career Member, Energy Frontier

2019 - 2022

- Participated as an Early Career member in Snowmass, the decadal community planning process (organized by the American Physical Society's Division of Particles and Fields) that shapes U.S. particle physics research priorities and informs the Department of Energy's funding strategy via the Particle Physics Project Prioritization Panel (P5) report.
- Co-authored a Snowmass white paper on anomalous quartic gauge couplings at a proposed muon collider, simulating signal and background collisions using MadGraph, Pythia, and Delphes within an effective field theory framework; results were submitted for publication in *Physical Review D* (arXiv:2203.08135).
- White paper findings directly informed Department of Energy (DOE) research priorities and funding recommendations for future particle collider initiatives, contributing evidence to the P5 process shaping the next decade of U.S. high-energy physics investment.

Teaching & Instruction Experience

University of Oklahoma

Graduate Teaching Assistant

Norman, OK

Aug. 2021 - Dec. 2024

- Provided instructional support and taught lab sections to diverse student groups, communicating complex scientific topics clearly to learners at different levels.
- Tutored undergraduates in General Education Physics I and II, reinforcing core concepts for non-major and introductory-level students.
- Taught astronomy lab sections for non-major undergraduates, guiding hands-on observational and data analysis exercises.
- Mentored Research Experiences for Undergraduates (REU) students in both high-energy physics and astrophysics research projects.
- Designed, reviewed, and evaluated scientific instructional materials, ensuring clarity for students with varied technical backgrounds.

The Carpentries

Certified Software, Data, & Library Carpentries Instructor

Dec. 2025 - Present

- Teach and assist with Software, Data, and Library Carpentry workshops covering Unix, Python, R, Git/GitHub version control, and Docker.
- Member of The Carpentries AI Working Groups and Community of Practice, supporting the development of AI workshops and ethical best practices for teaching AI to novice researchers.

Publications

Reach, W. T., Kilic, M., Lisse, C. M., Debes, J. H., von Hippel, T., **Azartash-Namin, B.**, Albert, L., Mullally, S. E., Mullally, F., Cracraft, M., Bernice, M., & Erickson, S. L. (2025). Composition of planetary debris around the white dwarf GD 362. *The Astrophysical Journal*, 994(2), 1-14. <https://doi.org/10.3847/1538-4357/ae11a9>

Blouin, S., Kilic, M., Albert, L., **Azartash-Namin, B.**, & Dufour, P. (2024). JWST resolves collision-induced absorption features in white dwarfs. *The Astrophysical Journal*, 976(2), 1-14. <https://doi.org/10.3847/1538-4357/ad863b>

Azartash-Namin, B., Engelhardt, A., Munshi, F., Keller, B. W., Brooks, A. M., Van Nest, J., Christensen, C. R., Quinn, T., & Wadsley, J. (2024). Bursting with feedback: The relationship between feedback model and bursty star formation histories in dwarf galaxies. *The Astrophysical Journal*, 970(1), 1-14. <https://doi.org/10.3847/1538-4357/ad49a5>

Abbott, B., Apyan, A., **Azartash-Namin, B.**, Balakrishnan, V., Berryhill, J., Hsu, S.-C., Jindariani, S., Kawale, M. P., Khoda, E. E., Parsons, R., Schuy, A., Strauss, M., Stupak, J., & Waits, C. (2023). Anomalous production of massive gauge boson pairs at muon colliders. *Physical Review D*, 108(9), 1-9. <https://doi.org/10.1103/PhysRevD.108.093009>

Belloni, A., Freitas, A., Tian, J., et al., including **Azartash-Namin, B.** (2022). Report of the topical group on electroweak precision physics and constraining new physics for Snowmass 2021. *arXiv preprint*. <https://doi.org/10.48550/arXiv.2209.08078>

Presentations & Panels

Edwards, J., **Azartash-Namin, B.**, & Rankin, D. (2026, April). *A librarian's approach to AI* [Conference presentation]. Oklahoma Library Association Annual Conference, Tulsa, OK, United States.

Azartash-Namin, B. (2026). *Research Data Management* [Invited Panelist]. OSCER Virtual Residency Workshop, University of Oklahoma.

Azartash-Namin, B. (2026). *AI in Libraries & Higher Education* [Invited Panelist]. Libraries at OU Day, University of Oklahoma.

Azartash-Namin, B. (2026, February). *Concepts of Data-Driven Visualization* [Presentation]. Love Data Week 2026.

Certifications & Professional Development

- Data Management & Sharing — Center for Open Science (June 2026)
- Accessibility in Data Visualization — Open Visualization Academy (January 2026)
- Information Design and Data Visualization — Open Visualization Academy (January 2026)
- Certified Software, Data, & Library Carpentries Instructor — The Carpentries (December 2025)

Service & Committee Membership

University Libraries

- **AI Action Team for Research Support and Student Success**, *Member*
- **UL Research Workshops Committee**: Answer data and AI questions as a resource to the committee's administrative planning for AI, Research, and Carpentries workshops.

- **AI Discussion Group 2026:** Co-led internal discussion series at University Libraries for faculty and staff over various AI topics.

University of Oklahoma — *University-Wide*

- **AI Education Working Group, Member.** Working group awarded the Oklahoma Online Excellence Award for the AI 101 Tutorial.

University of Oklahoma — *Dept. of Physics & Astronomy*

- **Women in Physics:** Led STEM diversity initiatives through strategic communication and community outreach as a member of WiP leadership.
- **Lunar Sooners (Astronomy Outreach Club), Member**

The Carpentries

- **AI Working Groups and Community of Practice, Member**

N-Body Shop

- **Code of Conduct Committee:** Led efforts in drafting and enforcing policies that promote inclusivity and ethical research practices.

Relevant Projects & Deliverables

Data Services & Research Support

- **AI Sandbox Pilot Project:** documentation, support materials, and user guides for LibreChat, LiteLLM, and Hugging Face + Nautilus, developed in collaboration with the DSDS team
- **Workshops:** “Concepts of Data-Driven Visualization” (Love Data Week 2026), “Programming with LLMs (APIs and Python)”, “AI Fundamentals for Researchers” (in progress)
- **Guides:** LibreChat User Guide, LiteLLM User Guide, Hugging Face + Nautilus User Guide, Data Services documentation updates (in progress)

AI Literacy Instruction & Resources

- **AI Lab (Course Re-Design), Center for Faculty Excellence:** Prompt Engineering, Literature Searches with AI, SIFT for AI
- **Workshops:** “Academic Integrity vs. Artificial Intelligence”, “Making AI Work for You”, “Critical Doing: How to Fact-Check Generative AI”, “Build-a-Bot: Custom Chatbots”
- **Tutorials:** “Fact-Checking ChatGPT”, “SIFT for AI” (AI 101 Course), “Escaping WorkSlop”
- **Guides:** “AI in the Library” LibGuide, “AI Library Instruction Guide”, “Prompt Guide for Research and Learning”, “LibreChat Prompt Library”, Frontier Model Learning Guides (Study/Learning Modes, Deep Research, Custom GPTs for ChatGPT, Claude, Gemini, Perplexity)
- **Templates:** “AI Scaffolding Assignment”, “Custom Chatbot Design Worksheet”

Skills

Consultation, Instruction & Outreach: One-on-one and embedded research consultations, translating technical concepts for non-technical audiences, workshop design and delivery, curriculum/instruction guide development, needs assessment and referral, cross-unit coordination (UL, CFE, DISC, OSCER), stakeholder engagement, public speaking, project coordination

Research Data Management: Research data lifecycle, data management plans, metadata documentation, data organization & preservation, discipline-specific and generalist data repositories, data standards (file formats, metadata schemas), research data sharing, reproducible workflows, open science practices

AI Ethics & Literacy Frameworks: ACRL CHOICE AI Literacy Framework, ACRL Framework for Information Literacy for Higher Education, responsible and ethical AI use in research and instruction, AI literacy pedagogy, WCAG 2.1 digital accessibility standards as applied to AI-generated content

Technical:

- **Programming & Development Tools:** Python, R, Jupyter, Git/GitHub, LaTeX, Linux/Command Line, Unix Shell, Docker
- **AI/LLM Infrastructure & Tools:** LibreChat, LiteLLM, commercial and open-source Large Language Models (LLMs), Machine Learning/Neural Networks, APIs/Model Context Protocols (MCPs), AWS Bedrock
- **Research Computing & Data Platforms:** Open Science Framework (OSF), National Research Platform (NRP), OSCER high-performance computing/GPU environments, LibApps (Springshare)

- **High-Energy Physics Software:** ROOT (C/C++), MadGraph, Pythia, Delphes, ATLAS Athena framework, Monte Carlo simulation, effective field theory
- **Astrophysics Software:** JWST data reduction pipeline, MESA (Modules for Experiments in Stellar Astrophysics), IRAF (Image Reduction and Analysis Facility), dwarf galaxy simulations
- **Hardware & Instrumentation:** Arduino/embedded systems, sensor integration & environmental monitoring, 3D printing/rapid prototyping, experimental design & hardware testing

Analytical:

- **Data Analysis & Visualization:** Data analysis, data visualization, data wrangling and transformation, quantitative analysis
- **Research & Information Skills:** Information retrieval and verification, scientific writing, digital accessibility